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ORIGINAL ARTICLE

Evaluation of Association of Allergic Rhinitis with Bronchial Asthma

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Abstract Allergic rhinitis represents a global health problem. It is a common disease worldwide affecting about 10–50 % of the population and its prevalence is increasing. Although allergic rhinitis is not a fatal disease, it alters the social life of patients, affecting learning performance and work productivity. Moreover, the costs incurred by allergic rhinitis are substantial. In recent years allergic rhinitis has been recognized to be an important risk factor for asthma. The concept of "One Airway, One Disease" was highlighted in the 'Allergic rhinitis and its Impact on Asthma 'guidelines and has arisen as a result of the now well-established link between the upper and lower airways. The aim of this study was to evaluate the association between allergic rhinitis and bronchial asthma by determining the incidence of bronchial asthma in patients of allergic rhinitis and the incidence of allergic rhinitis in patients of bronchial asthma. 83 diagnosed cases each of allergic rhinitis and bronchial asthma were recruited from patients attending Otorhinolaryngology and pulmonary department of the institute. All patients were subjected to detail ENT and pulmonary examination and investigated for nasal and bronchial allergy. In the allergic group, which consisted of 83 diagnosed patients of allergic rhinitis, 49 (59.03 %) were diagnosed to have bronchial asthma, whereas in the bronchial asthma group, which consisted of 83 diagnosed patients of bronchial asthma 61 (78.20 %) were diagnosed to have comorbid allergic rhinitis. It was observed that patients with allergic rhinitis were likely to develop bronchial asthma, and patients of allergic rhinitis should be evaluated for bronchial asthma, for early detection and treatment of the co morbid condition.

Keywords Allergic rhinitis · Bronchial asthma · Bronchial hyperresponsiveness

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Introduction

Allergic rhinitis (AR) represents a global health problem. It is a common disease worldwide affecting about 10–50 % of the population and its prevalence is increasing. Although allergic rhinitis is not a fatal disease, it alters the social life of patients, affecting learning performance and work productivity. Moreover, the costs incurred by allergic rhinitis are substantial. In recent years allergic rhinitis has been recognized to be an important risk factor for asthma [1]. The concept of "One Airway, One Disease" was highlighted in the 'Allergic rhinitis and its Impact on Asthma 'guidelines and has arisen as a result of the now well-established link between the upper and lower airways [2]. AR is defined as a symptomatic immunoglobulin E (IgE) mediated inflammation of the nasal mucosa. Symptoms of allergic rhinitis are reversible and include nasal congestion/obstruction, rhinorrhea, sneezing, pruritis, post nasal drip, chronic cough, throat clearing and conjunctivitis. Asthma is a chronic inflammatory disorder of the airways that results in reversible airway obstruction and bronchial hyperresponsiveness (BHR) to a variety of stimuli. Asthma and allergic rhinitis are the most common allergic airway diseases with epidemic proportions and are both inflammatory diseases of the airways. The similarities between AR and asthma in epidemiological and pathophysiologic features suggest that AR and asthma represent the same syndrome, the chronic allergic respiratory syndrome [3]. Many patients with asthma particularly those with allergic asthma also have AR. In adults, it has been shown that AR has a major impact on asthma morbidity, and that treatment of AR helps to improve asthma control [4]. Results of observational studies indicate that treating comorbid AR results in a lowered risk of asthma related hospitalizations and emergency visits. There is therefore evidence suggesting that comorbid allergic rhinitis is a marker of more difficult to control asthma and worsened asthma outcomes [5].

Materials and Methods

This study was conducted in the department of Otorhinolaryngology of a tertiary care teaching institute over a period of 12 months. 83 diagnosed cases of allergic rhinitis and 83 diagnosed cases of bronchial asthma were recruited from patients attending Otorhinolaryngology and pulmonary department of the institute. The data was recorded as per proforma after taking written informed consent and prior permission from institutional ethics committee. Patients below 18 years of age, present or former smokers, patients with extensive lung parenchymal disease, severe cardiac disease and patients on bronchodilators (within 24 h) or steroids (in the last 2–3 weeks) were excluded from the study. All patients were subjected to detailed case history, nose examination which included anterior rhinoscopy, functional nasal examination and diagnostic nasal endoscopy, chest examination which included ausculation, spirometry, x ray chest and haematological investigations—absolute eosinophil count and total IgE levels for diagnosis of allergic rhinitis and bronchial asthma. Data was tabulated and described in the form of percentages and ratios. Chi square test was used for statistical analysis and P value of <0.05 was considered significant.

Results

In the present study, it was observed that the age distribution of the patients ranged from 18 to 72 years. The maximum number of patients in both the groups were in the age group of 18–30 years i.e. 45 (54.21 %) and 35 (42.16 %) in the AR and BA group respectively, followed by 31–40 years with 19 (22.89 %) and 22 (28.20 %) patients in the AR and BA group respectively. The mean age was 32.17 ± 12.53 and 35.32 ± 13.39 years in the AR and BA group, respectively. Both AR and BA were more commonly seen in men. The male: female ratio in the AR group was 2.45:1 whereas in the BA group male: female ratio was 1.86:1. The number of patients from urban areas were more in both the AR and BA group i.e. 43 (51.80 %) and 54 (65.06 %), respectively.

Table 1 Chief complaints of the patients

Chief complaint	Allergic rhinitis (N = 83)	Bronchial asthma (N = 83)
Excessive sneezing	46 (55.42 %)	12 (14.45 %)
Rhinorrhea	38 (45.78 %)	11 (12.82 %)
Nasal obstruction	9 (10.84 %)	2 (2.40 %)
Breathlessness	3 (3.61 %)	71 (85.54 %)
Nasal itching	1 (1.20 %)	0
Cough	1 (1.20 %)	20 (24.09 %)

The chief complaints of the patient in both the AR and BA group is shown in Table 1. In the AR group the most common complaint of the patient was excessive sneezing seen in 46 (55.42 %) patients, whereas in the BA group breathlessness was the most common complaint, seen in 71 (85.54 %) patients.

Most common symptom of allergic rhinitis observed in both AR and BA group was excessive sneezing seen in 81 (97.59 %) and 60 (72.28 %) patients, respectively followed by rhinorrhea, which was present in 78 (93.97 %) and 58 (69.23 %) patients. Most common symptom of bronchial asthma was breathlessness experienced by 41 (49.39 %) and 77 (92.77 %) patients in AR and BA group, respectively. Dry cough was seen in 33 (39.75 %) and 65 (78.31 %) patients, followed by wheezing experienced by 25 (30.12 %) and 68(81.92 %) patient of AR and BA group, respectively. 32 (38.5 %) patients of allergic rhinitis experienced no symptoms of bronchial asthma.

In both the AR and BA groups dust was the most common allergen leading to aggravation of symptoms on exposure in 71 (85.54 %) and 47 (56.62 %) patients, respectively ($P < 0.05$). Exposure to smoke caused symptom aggravation in 34(40.96 %) and 26 (31.32 %) patients of AR and BA group, respectively ($P = 0.1962$). However, the third most common precipitating factor for symptoms aggravation was pollen 25 (30.12 %) in the AR group, whereas in the BA group change in weather 12 (14.45 %) was a more common factor as compared to pollen.

Signs of allergic rhinitis were present in 63 (75.9 %) and 55 (66.26 %) patients of AR and BA group, respectively. In both the groups, conjunctival hyperemia was seen in 49 (77.77 %) and 48 (87.27 %) patients followed by Allergic shiner's in 44 (69.84 %) and 31 (56.36) patients in AR and BA group, respectively. Darrier's lines were seen in 25 (39.68 %) and 12 (21.81 %) patients in AR and BA group, respectively.

On diagnostic nasal endoscopy, nasal polyposis was seen in 11 (13.25 %) and 7 (8.43 %) patients of AR and BA group, respectively. Polypoidal nasal mucosa was present in 28 (33.73 %) and 22 (26.50 %) patients of AR and BA group, respectively. On chest auscultation in the AR group 64 (77.10 %) patients and in the BA group 25 (30.12 %) patients had normal vesicular breath sounds (NVBS). In AR group 15 (18.07 %) patients had ronchi, and 4 (4.81 %) had ronchi and wheeze together, whereas, in the BA group most of the patients i.e. 41 (49.39 %) had ronchi on auscultation, 10 (14.45 %) patients had ronchi and wheeze together, 3 (3.61 %) patients had wheeze only and 1 (1.20 %) patient each had presence of crepts alone and crepts and wheeze together.

Raised AEC was seen in 41 (49.39 %) and 46 (55.42 %) patients in the AR and BA group, respectively ($P = 0.437$). Raised IgE values were found in 37 (44.57 %) and 40 (48.19 %) patients of AR and BA group, respectively ($P = 0.64$).

Obstructive pattern on spirometry was present 24 (28.91 %) and 60 (72.29 %) patients in the AR and BA group, respectively. In the AR group, 73 (87.95 %) patients were diagnosed with bronchial asthma, out of which 49 (59.03 %) had both symptoms and clinical signs and 24 (28.91 %) patients had obstructive changes on spirometry. 2 (4.08 %) patients in AR group with comorbid bronchial asthma neither had symptoms nor chest findings of bronchial asthma, but were found to have obstructive disease on spirometry. While in BA group, 70 (84.34 %) patients were diagnosed to have allergic rhinitis on the basis of symptoms and clinical evaluation.

Figure 1 demonstrates the pattern of onset of symptoms of both BA and AR in patient having concomitant disease. Majority of the patient in the AR group, (developed symptoms of BA after symptoms of allergic rhinitis (succeeding BA), followed by concomitant onset of symptoms of both diseases and rest of the patients developed symptoms of BA before the symptoms of AR (preceding BA). However in BA group majority of patients had developed succeeding BA symptoms followed by concomitant disease and preceding BA symptoms.

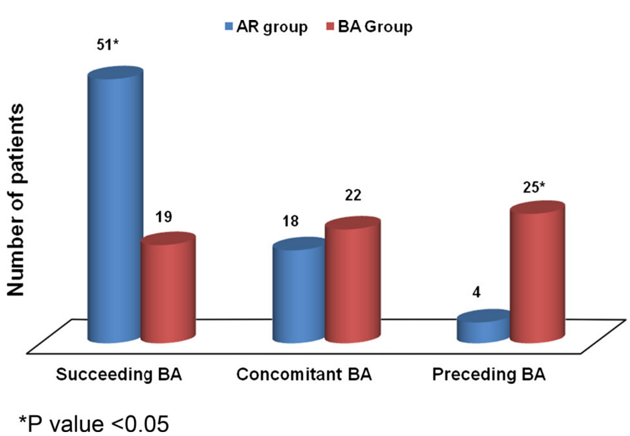

Fig. 1 Pattern of onset of symptoms of both BA and AR in patient having concomitant disease in both the groups. *$P < 0.05$

Table 2 Association between AR and BA with respect to time

	No. of patients in AR group with succeeding BA	No. of patients in BA group with preceding AR
<2 years	35*	5
>2 years	16	14

* $P = 0.001467$

Mean duration of development of bronchial asthma in patients of allergic rhinitis was 2.86 ± 3.52 years and development of allergic rhinitis in patients of bronchial asthma was 0.59 ± 4.93 years. Table 2 gives the duration of onset of symptom of AR prior to BA. The association between the two diseases, with respect to time was evaluated using Chi square test and was found to be significant ($P = 0.001467$).

Discussion

In the present study, it was observed that the age distribution of the cases ranged from 18 to 72 years. The mean age in the AR group and BA group was 32.17 ± 12.53 and 35.32 ± 13.39 years respectively. These findings were in accordance with one study, which included patients between 15 and 77 years with a mean age of 30.68 ± 10.65 years [6]. However, our study group differed from that of many other studies, since we excluded patients below 18 years of age, and most of the other studies were conducted on patients less than 18 years of age. One study included children between 7 and 9 years of age [7], whereas another study included patients aged between 16 and 75 years; and the mean age of the patients was 45.3 ± 16.2 years [8]. Since ours is a tertiary care centre, and children below 18 years, with airway diseases were treated by department of pediatrics, and neither by department of pulmonary medicine or otorhinolaryngology, hence this age group was excluded from our study. Male preponderance was seen in our study because the male population worked outdoors and were more prone to exposure to allergens such as dust, smoke, pollen and change in weather. Since the symptoms of AR and BA affected a male person's efficiency at work, they were more likely to seek medical help, than the female counterparts.

In our study, the number of cases from urban areas was more in both the AR and BA group. This was in accordance with the study, which showed that physician diagnosed asthma with comorbid allergic rhinitis was more common in urbanized population (68.5 %) [8].

Reason for increased number of patients from urban areas can be attributed to increased air pollution, urbanization and industrialization of the suburbs. The increased prevalence of these conditions in the urban population may also be attributed to hygiene hypothesis, which suggests that exposure to pathogens early in life protects against development of atopic phenotypes by contributing to immune system development.

In our study, in the AR group the most common chief complaint of the patient was excessive sneezing, followed by nasal obstruction (10.84 % patients), breathlessness (3.61 % patients), nasal itching and cough. Our findings differed from the study, where after sneezing seen in 94 % patients, unlike our study nasal obstruction (89.2 % patients) and nasal itching (87.1 % patients) were more common, as compared to rhinorrhea [6].

In the study conducted in Kolkata, the number of "blockers", i.e. patients with complaints of nasal obstruction was higher than the number of sneezers-runners, i.e. who had excessive sneezing and rhinorrhea [9].

However, in the BA group breathlessness was the most common chief complaint. These findings did not match with, one study where cough was the most common symptom, seen in 45.1 % patients, followed by dyspnea, seen in 44.4 % patients [6].

In our study population, patients in general did not recognize rhinitis or allergy as a disease and presented to the hospital, only when they developed troublesome symptoms of breathlessness or excessive sneezing or persistent rhinorrhea. Only one study, mentions that comorbid asthma and AR disrupted the ability to get a good night's sleep, concentrate at work/school, participate in leisure/sports and to enjoy social activities [10].

All the patients of both AR and BA presented to the hospital within 2 years of development of the symptoms. None of the related studies mentions the duration of symptoms, at the time of presentation to the hospital.

In the AR and BA group, majority of patients complained of aggravation of symptoms on exposure to dust followed by smoke and pollen. According to one study dust was the most common triggering factor seen in 39.5 % patients, followed by strong perfume odours and cold weather [11]. Another study reported grass pollen to be the most frequent allergen, followed by house dust mite [6]. A study revealed that, in comparison with their non allergic counterparts, individuals sensitized to pollen (a seasonal allergen) had a tenfold increased risk for developing asthma, whereas those who were sensitized to dust mite (a perennial allergen) had a 50-fold increased risk for developing asthma [12].In our study a significant number of patients were involved in agricultural activities and had exposure to dust and smoke while commuting on two wheelers, therefore they identified dust, smoke and pollen as the most common triggering factors in the aggravation of their symptoms.

Allergic conjunctivitis was the most common systemic manifestation seen in both the groups. Our findings were in concordance with the study which also reported eye itching and watering to be the most frequent eye symptoms seen in 82.1 % patients [6]. Allergic conjunctivitis was seen in 51.9 % patients in another study conducted in a medical centre in North Western Tanzania [11]. 61 % of adults experienced red, watery, itchy, or puffy eyes as reported a study published elsewhere [10].

In our study, in the AR group 24 (28.91 %) patients had obstructive pattern on spirometry, which was in accordance with the Spanish study conducted on patients of allergic rhinitis. 24 % of those, who did not report symptoms of bronchial asthma showed positive stress test, i.e. obstructive changes on spirometry, FEV1 > 15 % [13]. Another study reported pulmonary function abnormalities in 39.1 % patients, of which asthma was diagnosed in 26.1 % [14]. One study reported that in physician diagnosed patients of asthma, 63.9 % had allergic rhinitis, which was lower in comparison to our study where approximately 78 % patients of diagnosed BA had allergic rhinitis [8].

In our study mean AEC $\pm$ SD in the AR group was 0.50 ± 0.40, and in the BA group was 0.52 ± 0.34. However, AEC in the patients of allergic rhinitis with associated bronchial asthma was 0.54 ± 0.25, and those of bronchial asthma with associated allergic rhinitis were 0.52 ± 0.25. Our values differed from those reported in another study, which found a higher AEC in asthmatics, which was 0.70 (95 % CI 0.48–1.11) [15].The mean IgE levels in our study were found to be similar in both AR and BA groups IgE levels in the patients of allergic rhinitis with associated bronchial asthma were found to be slightly higher i.e. 274.09 ± 284.46, but were similar in patients of bronchial asthma with allergic rhinitis being 227.22 ± 240.34.

Prevalence of bronchial asthma in patients of allergic rhinitis in our study i.e. 59.03 % was lower in the study, where comorbid bronchial asthma was found in 50.2 % patients of allergic rhinitis [9]. Surveys have shown that approximately 60–80 % of children with asthma have symptoms of AR [9]. Presence of allergic rhinitis in patients of bronchial asthma which was 78.20 % of our study matched the above mentioned statement. This also coincided with the findings of two other studies which observed comorbid AR and asthma in 79 % of patients [10].

In a study conducted in Karnataka, it was found that 700 out of 1141 subjects had allergic rhinitis for varying intervals before developing asthma. The study also reported that rhinitis often preceded asthma and a high proportion of patients, both children and adults, developed asthma within 2 years after the onset of rhinitis [16]. Our study did not report a similar finding since we had a much smaller number of subjects and time duration was a limitation and we did not follow up patients for more than 1 year. However other authors have also mentioned that diagnosis of allergic rhinitis often precedes that of asthma [17]. In our study we found a significant association between the developments of the two conditions with respect to time.

Conclusion

Our study revealed that 59 % of allergic rhinitis patients had bronchial asthma and similarly 78 % of bronchial asthma patients had comorbid allergic rhinitis. In the allergic rhinitis patients having comorbid bronchial asthma 70 % patients developed subsequent bronchial asthma. Whereas, in the bronchial asthma group having comorbid allergic rhinitis only 29 % patient had preceding allergic rhinitis symptoms whereas 33 % had concomitant symptoms. It was found that the patients with allergic rhinitis were likely to develop bronchial asthma after 2 years and this association between the two diseases with respect to time was found to be highly significant.

Hence, it was observed that patients with allergic rhinitis were likely to develop bronchial asthma, and patients of allergic rhinitis should be evaluated for bronchial asthma, for early detection and treatment of the co morbid condition.

Compliance with Ethical Standards

Conflict of interest None.

Ethical Approval All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

Informed Consent Informed consent was obtained from all individual participants included in the study.

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